A3-PS003-TC307

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

Traffic flow prediction helps estimate future traffic conditions and improve traffic management. However, accurate prediction is challenging due to changing traffic patterns and congestion. Frequent data transmission between sensors and prediction systems also increases communication costs and network load.

Key Issues:

- Difficulty in predicting traffic flow accurately.

- High communication cost due to frequent data transmission.

- Delays or incomplete traffic data.

- Balancing prediction accuracy and communication efficiency.

The main objective is to develop an efficient traffic flow prediction approach that maintains accuracy while minimizing communication costs.

Background

Real-time traffic prediction is essential for smart cities, but transmitting raw sensor data to centralized cloud servers increases bandwidth usage and creates privacy risks. Federated Learning addresses these challenges by training AI models locally at edge nodes and sharing only model updates. However, frequent synchronization can still cause network congestion and resource overhead.

This project aims to minimize communication costs while maintaining high prediction accuracy using techniques such as model compression, quantization, and adaptive communication frequency. By evaluating communication overhead, prediction accuracy, and latency under varying network conditions, the proposed approach seeks to deliver privacy-preserving, resource-efficient, and scalable traffic forecasting for real-world smart cities.

Literature & RFC Review

Existing research identifies communication overhead as a major challenge in Federated Learning. FedAvg is a widely used baseline algorithm, while techniques such as model compression, gradient quantization, and adaptive client selection can reduce communication costs but may affect prediction accuracy. RFC 9000 defines QUIC, a secure transport protocol supporting multiplexed communication and low-latency connections. However, balancing communication efficiency, prediction accuracy, and real-time performance remains a key challenge in intelligent transportation systems. This project aims to evaluate these trade-offs and identify an efficient, scalable approach for real-time traffic prediction.

Proposed Solution

We propose a lightweight, decentralized framework for privacy-preserving traffic prediction with reduced communication overhead. Spatial-temporal models, such as ST-GCN or LSTM, will process traffic data locally at edge nodes, avoiding the need to transmit raw data to a central server. To reduce communication costs during federated aggregation, the framework will use adaptive gradient sparsification, model compression, and event-triggered updates based on significant traffic changes. The system will evaluate prediction accuracy, communication bandwidth, and transmission latency to identify an efficient balance between model performance and network resource consumption. This approach aims to support scalable, privacy-aware traffic forecasting for intelligent transportation systems.

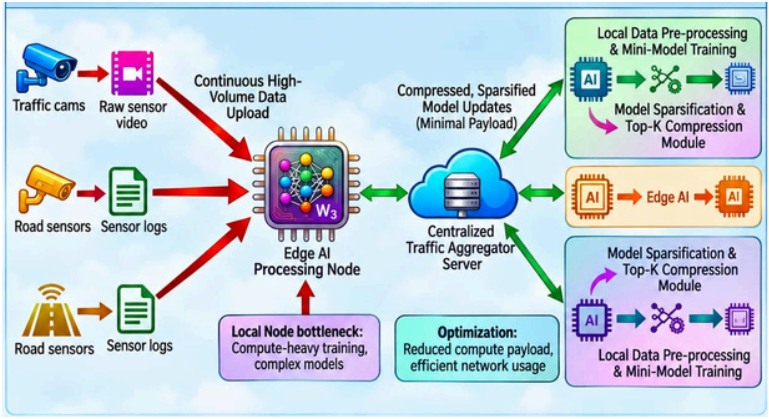
Optimization Proposed

To minimize communication overhead in federated traffic forecasting, we introduce three key enhancements: Gradient Sparsification, Model Quantization (8-bit), and Event-Driven Aggregation. By sending compressed parameter updates only when traffic patterns shift, we drastically cut bandwidth consumption while retaining top-tier prediction accuracy.

Parameter/Metric	**Baseline Federated Approach (Before)**	Our Optimized Architecture (After)
Data Payload Size	Full Model Updates (~100% data transfer)	70-80% Reduced Payload (Quantized & Sparse)
Aggregation Frequency	Fixed Periodic Rounds	Event- Driven Dynamic Sync
Network Bandwidth Usage	High Network Strain & Latency	Minimal Bandwidth & Low Latency
Prediction Accuracy	High (~92%)	Preserved High Accuracy (~91.5-93%)

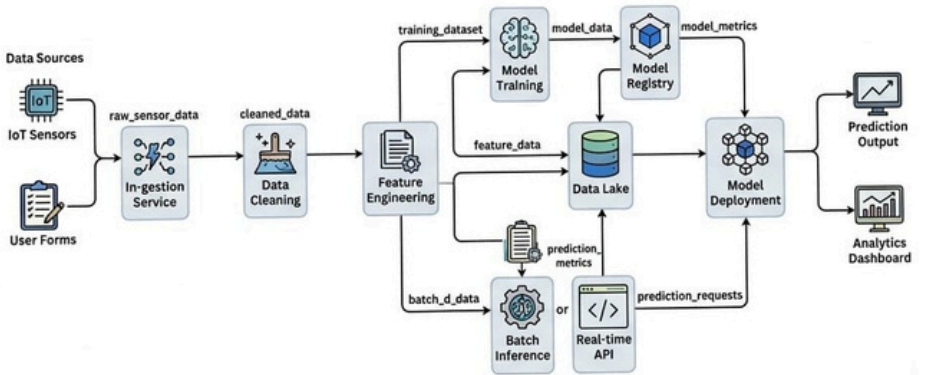
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.



Deployment Timeline

Deployment Timeline and Methodology

Phase 1 — Node Preparation & Data Hygiene (Weeks 1–2): Configure simulated edge nodes, prepare spatial-temporal traffic datasets, and establish privacy-preserving data handling procedures.

Phase 2 — Local Intelligence & Compression (Weeks 3–4): Implement local traffic prediction models and integrate Top-K gradient sparsification and quantization to reduce communication payloads.

Phase 3 — Federated Testing & Optimization (Weeks 5–6): Test federated aggregation under simulated traffic and network conditions, measuring bandwidth usage, communication latency, and prediction accuracy.

Phase 4 — Pilot Readiness & Robustness Evaluation (Weeks 7–8): Evaluate performance during simulated rush-hour surges, assess system reliability and security, and prepare the framework for potential real-world pilot deployment.

Bibliography I

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[4] S. Wang et al., “Adaptive Federated Learning in Resource Constrained Edge Computing Systems,” relevant research on communication-efficient federated learning and edge computing.

Thank You for Your Attention!